

IDENTIFICATION AND AUTHENTICATION	IA-1	IA-1	IDENTIFICATION AND AUTHENTICATION POLICY AND PROCEDURES	The organization:
IDENTIFICATION AND AUTHENTICATION	IA-1	IA-1a	IDENTIFICATION AND AUTHENTICATION POLICY AND PROCEDURES	Develops, documents, and disseminates to [Assignment: organization-defined personnel or roles]:
IDENTIFICATION AND AUTHENTICATION	IA-1	IA-1a.1	IDENTIFICATION AND AUTHENTICATION POLICY AND PROCEDURES	An identification and authentication policy that addresses purpose, scope, roles, responsibilities, management commitment, coordination among organizational entities, and compliance; and
IDENTIFICATION AND AUTHENTICATION	IA-1	IA-1a.2	IDENTIFICATION AND AUTHENTICATION POLICY AND PROCEDURES	Procedures to facilitate the implementation of the identification and authentication policy and associated identification and authentication controls; and
IDENTIFICATION AND AUTHENTICATION	IA-1	IA-1b	IDENTIFICATION AND AUTHENTICATION POLICY AND PROCEDURES	Reviews and updates the current:
IDENTIFICATION AND AUTHENTICATION	IA-1	IA-1b.1	IDENTIFICATION AND AUTHENTICATION POLICY AND PROCEDURES	Identification and authentication policy [Assignment: organization-defined frequency]; and

IDENTIFICATION AND AUTHENTICATION	IA-1	IA-1b.2	IDENTIFICATION AND AUTHENTICATION POLICY AND PROCEDURES	Identification and authentication procedures [Assignment: organization-defined frequency].
IDENTIFICATION AND AUTHENTICATION	IA-2	IA-2	IDENTIFICATION AND AUTHENTICATION (ORGANIZATIONAL USERS)	The information system uniquely identifies and authenticates organizational users (or processes acting on behalf of organizational users).
IDENTIFICATION AND AUTHENTICATION	IA-2 (1)	IA-2 (1)	NETWORK ACCESS TO PRIVILEGED ACCOUNTS	The information system implements multifactor authentication for network access to privileged accounts.

IDENTIFICATION AND AUTHENTICATION	IA-2 (2)	IA-2 (2)	NETWORK ACCESS TO NON-PRIVILEGED ACCOUNTS	The information system implements multifactor authentication for network access to non-privileged accounts.
IDENTIFICATION AND AUTHENTICATION	IA-2 (3)	IA-2 (3)	LOCAL ACCESS TO PRIVILEGED ACCOUNTS	The information system implements multifactor authentication for local access to privileged accounts.
IDENTIFICATION AND AUTHENTICATION	IA-2 (4)	IA-2 (4)	LOCAL ACCESS TO NON-PRIVILEGED ACCOUNTS	The information system implements multifactor authentication for local access to non-privileged accounts.

IDENTIFICATION AND AUTHENTICATION	IA-2 (5)	IA-2 (5)	GROUP AUTHENTICATION	The organization requires individuals to be authenticated with an individual authenticator when a group authenticator is employed.
IDENTIFICATION AND AUTHENTICATION	IA-2 (6)	IA-2 (6)	NETWORK ACCESS TO PRIVILEGED ACCOUNTS - SEPARATE DEVICE	The information system implements multifactor authentication for network access to privileged accounts such that one of the factors is provided by a device separate from the system gaining access and the device meets [Assignment: organization-defined strength of mechanism requirements].
IDENTIFICATION AND AUTHENTICATION	IA-2 (7)	IA-2 (7)	NETWORK ACCESS TO NON-PRIVILEGED ACCOUNTS - SEPARATE DEVICE	The information system implements multifactor authentication for network access to non-privileged accounts such that one of the factors is provided by a device separate from the system gaining access and the device meets [Assignment: organization-defined strength of mechanism requirements].
IDENTIFICATION AND AUTHENTICATION	IA-2 (8)	IA-2 (8)	NETWORK ACCESS TO PRIVILEGED ACCOUNTS - REPLAY RESISTANT	The information system implements replay-resistant authentication mechanisms for network access to privileged accounts.
IDENTIFICATION AND AUTHENTICATION	IA-2 (9)	IA-2 (9)	NETWORK ACCESS TO NON-PRIVILEGED ACCOUNTS - REPLAY RESISTANT	The information system implements replay-resistant authentication mechanisms for network access to non-privileged accounts.
IDENTIFICATION AND AUTHENTICATION	IA-2 (10)	IA-2 (10)	SINGLE SIGN-ON	The information system provides a single sign-on capability for [Assignment: organization-defined information system accounts and services].

IDENTIFICATION AND AUTHENTICATION	IA-2 (11)	IA-2 (11)	REMOTE ACCESS - SEPARATE DEVICE	The information system implements multifactor authentication for remote access to privileged and non-privileged accounts such that one of the factors is provided by a device separate from the system gaining access and the device meets [Assignment: organization-defined strength of mechanism requirements].
IDENTIFICATION AND AUTHENTICATION	IA-2 (12)	IA-2 (12)	ACCEPTANCE OF PIV CREDENTIALS	The information system accepts and electronically verifies Personal Identity Verification (PIV) credentials.
IDENTIFICATION AND AUTHENTICATION	IA-2 (13)	IA-2 (13)	OUT-OF-BAND AUTHENTICATION	The information system implements [Assignment: organization-defined out-of-band authentication] under [Assignment: organization-defined conditions].
IDENTIFICATION AND AUTHENTICATION	IA-3	IA-3	DEVICE IDENTIFICATION AND AUTHENTICATION	The information system uniquely identifies and authenticates [Assignment: organization-defined specific and/or types of devices] before establishing a [Selection (one or more): local; remote; network] connection.
IDENTIFICATION AND AUTHENTICATION	IA-3 (1)	IA-3 (1)	CRYPTOGRAPHIC BIDIRECTIONAL AUTHENTICATION	The information system authenticates [Assignment: organization-defined specific devices and/or types of devices] before establishing [Selection (one or more): local; remote; network] connection using bidirectional authentication that is cryptographically based.

IDENTIFICATION AND AUTHENTICATION	IA-3 (2)	IA-3 (2)	CRYPTOGRAPHIC BIDIRECTIONAL NETWORK AUTHENTICATION	[Withdrawn: Incorporated into IA-3 (1)].
IDENTIFICATION AND AUTHENTICATION	IA-3 (3)	IA-3 (3)	DYNAMIC ADDRESS ALLOCATION	The organization:
IDENTIFICATION AND AUTHENTICATION	IA-3 (3)	IA-3 (3)(a)	DYNAMIC ADDRESS ALLOCATION	Standardizes dynamic address allocation lease information and the lease duration assigned to devices in accordance with [Assignment: organization-defined lease information and lease duration]; and
IDENTIFICATION AND AUTHENTICATION	IA-3 (3)	IA-3 (3)(b)	DYNAMIC ADDRESS ALLOCATION	Audits lease information when assigned to a device.
IDENTIFICATION AND AUTHENTICATION	IA-3 (4)	IA-3 (4)	DEVICE ATTESTATION	The organization ensures that device identification and authentication based on attestation is handled by [Assignment: organization-defined configuration management process].
IDENTIFICATION AND AUTHENTICATION	IA-4	IA-4	IDENTIFIER MANAGEMENT	The organization manages information system identifiers by:
IDENTIFICATION AND AUTHENTICATION	IA-4	IA-4a	IDENTIFIER MANAGEMENT	Receiving authorization from [Assignment: organization-defined personnel or roles] to assign an individual, group, role, or device identifier;

IDENTIFICATION AND AUTHENTICATION	IA-4	IA-4b	IDENTIFIER MANAGEMENT	Selecting an identifier that identifies an individual, group, role, or device;
IDENTIFICATION AND AUTHENTICATION	IA-4	IA-4c	IDENTIFIER MANAGEMENT	Assigning the identifier to the intended individual, group, role, or device;
IDENTIFICATION AND AUTHENTICATION	IA-4	IA-4d	IDENTIFIER MANAGEMENT	Preventing reuse of identifiers for [Assignment: organization-defined time period]; and
IDENTIFICATION AND AUTHENTICATION	IA-4	IA-4e	IDENTIFIER MANAGEMENT	Disabling the identifier after [Assignment: organization-defined time period of inactivity].
IDENTIFICATION AND AUTHENTICATION	IA-4 (1)	IA-4 (1)	PROHIBIT ACCOUNT IDENTIFIERS AS PUBLIC IDENTIFIERS	The organization prohibits the use of information system account identifiers that are the same as public identifiers for individual electronic mail accounts.
IDENTIFICATION AND AUTHENTICATION	IA-4 (2)	IA-4 (2)	SUPERVISOR AUTHORIZATION	The organization requires that the registration process to receive an individual identifier includes supervisor authorization.
IDENTIFICATION AND AUTHENTICATION	IA-4 (3)	IA-4 (3)	MULTIPLE FORMS OF CERTIFICATION	The organization requires multiple forms of certification of individual identification be presented to the registration authority.

IDENTIFICATION AND AUTHENTICATION	IA-4 (4)	IA-4 (4)	IDENTIFY USER STATUS	The organization manages individual identifiers by uniquely identifying each individual as [Assignment: organization-defined characteristic identifying individual status].
IDENTIFICATION AND AUTHENTICATION	IA-4 (5)	IA-4 (5)	DYNAMIC MANAGEMENT	The information system dynamically manages identifiers.
IDENTIFICATION AND AUTHENTICATION	IA-4 (6)	IA-4 (6)	CROSS- ORGANIZATION MANAGEMENT	The organization coordinates with [Assignment: organization-defined external organizations] for cross-organization management of identifiers.
IDENTIFICATION AND AUTHENTICATION	IA-4 (7)	IA-4 (7)	IN-PERSON REGISTRATION	The organization requires that the registration process to receive an individual identifier be conducted in person before a designated registration authority.
IDENTIFICATION AND AUTHENTICATION	IA-5	IA-5	AUTHENTICATOR MANAGEMENT	The organization manages information system authenticators by:

IDENTIFICATION AND AUTHENTICATION	IA-5	IA-5a	AUTHENTICATOR MANAGEMENT	Verifying, as part of the initial authenticator distribution, the identity of the individual, group, role, or device receiving the authenticator;
IDENTIFICATION AND AUTHENTICATION	IA-5	IA-5b	AUTHENTICATOR MANAGEMENT	Establishing initial authenticator content for authenticators defined by the organization;
IDENTIFICATION AND AUTHENTICATION	IA-5	IA-5c	AUTHENTICATOR MANAGEMENT	Ensuring that authenticators have sufficient strength of mechanism for their intended use;

IDENTIFICATION AND AUTHENTICATION	IA-5	IA-5d	AUTHENTICATOR MANAGEMENT	Establishing and implementing administrative procedures for initial authenticator distribution, for lost/compromised or damaged authenticators, and for revoking authenticators;
IDENTIFICATION AND AUTHENTICATION	IA-5	IA-5e	AUTHENTICATOR MANAGEMENT	Changing default content of authenticators prior to information system installation;
IDENTIFICATION AND AUTHENTICATION	IA-5	IA-5f	AUTHENTICATOR MANAGEMENT	Establishing minimum and maximum lifetime restrictions and reuse conditions for authenticators;

IDENTIFICATION AND AUTHENTICATION	IA-5	IA-5g	AUTHENTICATOR MANAGEMENT	Changing/refreshing authenticators [Assignment: organization-defined time period by authenticator type];
IDENTIFICATION AND AUTHENTICATION	IA-5	IA-5h	AUTHENTICATOR MANAGEMENT	Protecting authenticator content from unauthorized disclosure and modification;
IDENTIFICATION AND AUTHENTICATION	IA-5	IA-5i	AUTHENTICATOR MANAGEMENT	Requiring individuals to take, and having devices implement, specific security safeguards to protect authenticators; and

IDENTIFICATION AND AUTHENTICATION	IA-5	IA-5j	AUTHENTICATOR MANAGEMENT	Changing authenticators for group/role accounts when membership to those accounts changes.
IDENTIFICATION AND AUTHENTICATION	IA-5 (1)	IA-5 (1)	PASSWORD-BASED AUTHENTICATION	The information system, for password-based authentication:
IDENTIFICATION AND AUTHENTICATION	IA-5 (1)	IA-5 (1)(a)	PASSWORD-BASED AUTHENTICATION	Enforces minimum password complexity of [Assignment: organization-defined requirements for case sensitivity, number of characters, mix of upper-case letters, lower-case letters, numbers, and special characters, including minimum requirements for each type];

IDENTIFICATION AND AUTHENTICATION	IA-5 (1)	IA-5 (1)(b)	PASSWORD-BASED AUTHENTICATION	Enforces at least the following number of changed characters when new passwords are created: [Assignment: organization-defined number];
IDENTIFICATION AND AUTHENTICATION	IA-5 (1)	IA-5 (1)(c)	PASSWORD-BASED AUTHENTICATION	Stores and transmits only cryptographically-protected passwords;
IDENTIFICATION AND AUTHENTICATION	IA-5 (1)	IA-5 (1)(d)	PASSWORD-BASED AUTHENTICATION	Enforces password minimum and maximum lifetime restrictions of [Assignment: organization-defined numbers for lifetime minimum, lifetime maximum];

IDENTIFICATION AND AUTHENTICATION	IA-5 (1)	IA-5 (1)(e)	PASSWORD-BASED AUTHENTICATION	Prohibits password reuse for [Assignment: organization-defined number] generations; and
IDENTIFICATION AND AUTHENTICATION	IA-5 (1)	IA-5 (1)(f)	PASSWORD-BASED AUTHENTICATION	Allows the use of a temporary password for system logons with an immediate change to a permanent password.
IDENTIFICATION AND AUTHENTICATION	IA-5 (2)	IA-5 (2)	PKI-BASED AUTHENTICATION	The information system, for PKI-based authentication:
IDENTIFICATION AND AUTHENTICATION	IA-5 (2)	IA-5 (2)(a)	PKI-BASED AUTHENTICATION	Validates certifications by constructing and verifying a certification path to an accepted trust anchor including checking certificate status information;

IDENTIFICATION AND AUTHENTICATION	IA-5 (2)	IA-5 (2)(b)	PKI-BASED AUTHENTICATION	Enforces authorized access to the corresponding private key;
IDENTIFICATION AND AUTHENTICATION	IA-5 (2)	IA-5 (2)(c)	PKI-BASED AUTHENTICATION	Maps the authenticated identity to the account of the individual or group; and
IDENTIFICATION AND AUTHENTICATION	IA-5 (2)	IA-5 (2)(d)	PKI-BASED AUTHENTICATION	Implements a local cache of revocation data to support path discovery and validation in case of inability to access revocation information via the network.
IDENTIFICATION AND AUTHENTICATION	IA-5 (3)	IA-5 (3)	IN-PERSON OR TRUSTED THIRD- PARTY REGISTRATION	The organization requires that the registration process to receive [Assignment: organization-defined types of and/or specific authenticators] be conducted [Selection: in person; by a trusted third party] before [Assignment: organization-defined registration authority] with authorization by [Assignment: organization-defined personnel or roles].
IDENTIFICATION AND AUTHENTICATION	IA-5 (4)	IA-5 (4)	AUTOMATED SUPPORT FOR PASSWORD STRENGTH DETERMINATION	The organization employs automated tools to determine if password authenticators are sufficiently strong to satisfy [Assignment: organization-defined requirements].
IDENTIFICATION AND AUTHENTICATION	IA-5 (5)	IA-5 (5)	CHANGE AUTHENTICATORS PRIOR TO DELIVERY	The organization requires developers/installers of information system components to provide unique authenticators or change default authenticators prior to delivery/installation.

IDENTIFICATION AND AUTHENTICATION	IA-5 (6)	IA-5 (6)	PROTECTION OF AUTHENTICATORS	The organization protects authenticators commensurate with the security category of the information to which use of the authenticator permits access.
IDENTIFICATION AND AUTHENTICATION	IA-5 (7)	IA-5 (7)	NO EMBEDDED UNENCRYPTED STATIC AUTHENTICATORS	The organization ensures that unencrypted static authenticators are not embedded in applications or access scripts or stored on function keys.
IDENTIFICATION AND AUTHENTICATION	IA-5 (8)	IA-5 (8)	MULTIPLE INFORMATION SYSTEM ACCOUNTS	The organization implements [Assignment: organization-defined security safeguards] to manage the risk of compromise due to individuals having accounts on multiple information systems.
IDENTIFICATION AND AUTHENTICATION	IA-5 (9)	IA-5 (9)	CROSS- ORGANIZATION CREDENTIAL MANAGEMENT	The organization coordinates with [Assignment: organization-defined external organizations] for cross-organization management of credentials.
IDENTIFICATION AND AUTHENTICATION	IA-5 (10)	IA-5 (10)	DYNAMIC CREDENTIAL ASSOCIATION	The information system dynamically provisions identities.
IDENTIFICATION AND AUTHENTICATION	IA-5 (11)	IA-5 (11)	HARDWARE TOKEN- BASED AUTHENTICATION	The information system, for hardware token-based authentication, employs mechanisms that satisfy [Assignment: organization-defined token quality requirements].

IDENTIFICATION AND AUTHENTICATION	IA-5 (12)	IA-5 (12)	BIOMETRIC-BASED AUTHENTICATION	The information system, for biometric-based authentication, employs mechanisms that satisfy [Assignment: organization-defined biometric quality requirements].
IDENTIFICATION AND AUTHENTICATION	IA-5 (13)	IA-5 (13)	EXPIRATION OF CACHED AUTHENTICATORS	The information system prohibits the use of cached authenticators after [Assignment: organization-defined time period].
IDENTIFICATION AND AUTHENTICATION	IA-5 (14)	IA-5 (14)	MANAGING CONTENT OF PKI TRUST STORES	The organization, for PKI-based authentication, employs a deliberate organization-wide methodology for managing the content of PKI trust stores installed across all platforms including networks, operating systems, browsers, and applications.
IDENTIFICATION AND AUTHENTICATION	IA-5 (15)	IA-5 (15)	FICAM-APPROVED PRODUCTS AND SERVICES	The organization uses only FICAM-approved path discovery and validation products and services.
IDENTIFICATION AND AUTHENTICATION	IA-6	IA-6	AUTHENTICATOR FEEDBACK	The information system obscures feedback of authentication information during the authentication process to protect the information from possible exploitation/use by unauthorized individuals.
IDENTIFICATION AND AUTHENTICATION	IA-7	IA-7	CRYPTOGRAPHIC MODULE AUTHENTICATION	The information system implements mechanisms for authentication to a cryptographic module that meet the requirements of applicable federal laws, Executive Orders, directives, policies, regulations, standards, and guidance for such authentication.

IDENTIFICATION AND AUTHENTICATION	IA-8	IA-8	IDENTIFICATION AND AUTHENTICATION (NON- ORGANIZATIONAL USERS)	The information system uniquely identifies and authenticates non-organizational users (or processes acting on behalf of non-organizational users).
IDENTIFICATION AND AUTHENTICATION	IA-8 (1)	IA-8 (1)	ACCEPTANCE OF PIV CREDENTIALS FROM OTHER AGENCIES	The information system accepts and electronically verifies Personal Identity Verification (PIV) credentials from other federal agencies.
IDENTIFICATION AND AUTHENTICATION	IA-8 (2)	IA-8 (2)	ACCEPTANCE OF THIRD-PARTY CREDENTIALS	The information system accepts only FICAM-approved third-party credentials.
IDENTIFICATION AND AUTHENTICATION	IA-8 (3)	IA-8 (3)	USE OF FICAM- APPROVED PRODUCTS	The organization employs only FICAM-approved information system components in [Assignment: organization-defined information systems] to accept third-party credentials.
IDENTIFICATION AND AUTHENTICATION	IA-8 (4)	IA-8 (4)	USE OF FICAM- ISSUED PROFILES	The information system conforms to FICAM-issued profiles.
IDENTIFICATION AND AUTHENTICATION	IA-8 (5)	IA-8 (5)	ACCEPTANCE OF PIV- I CREDENTIALS	The information system accepts and electronically verifies Personal Identity Verification-I (PIV-I) credentials.

IDENTIFICATION AND AUTHENTICATION	IA-9	IA-9	SERVICE IDENTIFICATION AND AUTHENTICATION	The organization identifies and authenticates [Assignment: organization-defined information system services] using [Assignment: organization-defined security safeguards].
IDENTIFICATION AND AUTHENTICATION	IA-9 (1)	IA-9 (1)	INFORMATION EXCHANGE	The organization ensures that service providers receive, validate, and transmit identification and authentication information.
IDENTIFICATION AND AUTHENTICATION	IA-9 (2)	IA-9 (2)	TRANSMISSION OF DECISIONS	The organization ensures that identification and authentication decisions are transmitted between [Assignment: organization-defined services] consistent with organizational policies.
IDENTIFICATION AND AUTHENTICATION	IA-10	IA-10	ADAPTIVE IDENTIFICATION AND AUTHENTICATION	The organization requires that individuals accessing the information system employ [Assignment: organization-defined supplemental authentication techniques or mechanisms] under specific [Assignment: organization-defined circumstances or situations].
IDENTIFICATION AND AUTHENTICATION	IA-11	IA-11	RE-AUTHENTICATION	The organization requires users and devices to re-authenticate when [Assignment: organization-defined circumstances or situations requiring re-authentication].